

2. Using shielded cables
3. Using EMI gaskets to cover discontinuities in metallic enclosure at the places of connectors and cable entry points
4. Using EMI filters

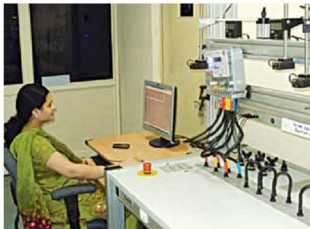
Electrostatic discharge. When certain types of material come into contact and separate, electrons pass back and forth between them. One material gets positively charged and the other negatively charged, for example, when walking on a carpet or rubbing a plastic comb against dry hair. This phenomenon is called electrostatic discharge (ESD). The amount of charge and voltage developed depends on the materials, area of contact, speed of separation and relative humidity. The voltage developed could be as high as 20,000V.

Static voltage is generated in work areas by people during their normal movement. This voltage can cause malfunction of parts, assemblies and equipment or even damage them. Certain devices and technologies are more sensitive to ESD than others. Components that include field-effect transistors (FETs) such as complementary metal-oxide semiconductor (CMOS), high-frequency parts produced by Schottky barrier process and micro-circuits that utilise small active junctions are sensitive to ESD.

Failures caused by ESD could be intermittent, catastrophic or latent. In case of latent defect, the damaged device remains operational and fails some time later in its use.

General protection techniques to be followed to provide immunity against ESD are:

1. Select components that have built-in protection against ESD, or build protection in circuit design by using diodes, silicon-controlled rectifiers, metal-oxide varistors, etc.
2. Enclose the circuit in a metallic



System entry test bench

box and establish wire connection from the box to ground. Any static discharge will transfer from the box through the wire connection to ground. If a plastic enclosure is used, coat its interior with a conductive paint.

3. During production and assembly, components and sub-assemblies should be handled such that these are not exposed to ESD. This can be achieved by doing assembly in an ESD-controlled work area using ESD work benches with ESD straps and seats. These ensure that the operator or user is grounded.

4. Personnel doing assembly should wear ESD clothing to minimise ESD generation.

5. All components and assemblies should be stored in an ESD-controlled area.

6. All components and assemblies should be handled and transported in conductive or ESD-protective bags.

Others. Manufacturability, testability and maintainability aspects should also be considered during PCB, module and package design. Specialists in all fields should be involved from initial stages of design, in addition to core functional designers. This is called concurrent engineering. Get the design reviewed by experts in relevant fields with respect to all the given aspects.

Documentation

Documents need to be prepared that give information on materials and components to be used, as well as electrical

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